

Lecture 21: Diffusion models for text

PSYC 51.17: Models of language and communication

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Learning objectives

By the end of this lecture, you will be able to...

1. Explain how **diffusion models** generate content by iteratively refining noise into signal
2. Describe the **forward** and **reverse** processes in continuous diffusion (foundation for Lectures 22–23)
3. Explain how **discrete diffusion** adapts the framework for text by replacing Gaussian noise with **masking**
4. Compare **autoregressive** and **diffusion-based** text generation and articulate their tradeoffs
5. Build intuition for why iterative refinement is a powerful paradigm for language generation

Announcements

Assignment 4 due today!

Assignment 4 (Customer Service Chatbot) is due **today at 11:59 PM EST.**

Final project and optional assignment

- **Final Project** has been released! Due **March 9, 11:59 PM EST**
- **Assignment 5** (Build GPT) is now available as **optional extra credit**

Final project

Your LLM Research Capstone

Work in **teams of 2-3(ish)** to build an ambitious, open-ended project over **Weeks 7-10**.

Explore novel applications, replicate and extend published research, build innovative systems, or evaluate LLM capabilities.

Timeline

- **Weeks 7–8:** Form teams and brainstorm ideas
- **Week 9:** Core development and experiments
- **Week 10:** Final submission + in-class presentations (**March 9, 11:59 PM EST**)

Get started

- Full instructions: [Final Project](#)
- Accept the assignment: [GitHub Classroom](#)
- 20+ project ideas on the assignment page to inspire you!

Why diffusion models?

A different approach to generation

Every language model we've studied so far — from ELIZA to GPT — generates text **one token at a time**, left to right. Diffusion models take a fundamentally different approach: start with **pure noise** and iteratively refine it into a coherent output, generating **all positions simultaneously**.

The core analogy

Imagine writing an essay by first typing random letters into every position, then making many editing passes — each pass fixing more of the text until a polished essay emerges. That's diffusion. It sounds absurd, but it works remarkably well — and for some tasks, it's **better** than left-to-right generation.

Two paradigms for generation

Autoregressive vs diffusion

	Autoregressive (GPT)	Diffusion
Generation order	Left to right, one token at a time	All positions at once, refined iteratively
Analogy	Speaking a sentence word by word	Editing a rough draft into a polished one
Key strength	Simple, proven at scale	Bidirectional context, flexible editing
Key weakness	Can't "go back" — early errors propagate	Requires many refinement steps

Why this matters for language

When you write, you don't commit to each word in sequence. You draft, revise, restructure. Diffusion models formalize this **iterative refinement** process — and recent results show they can match autoregressive models at the scale of billions of parameters ([LLaDA](#); [Nie et al., 2025](#)).

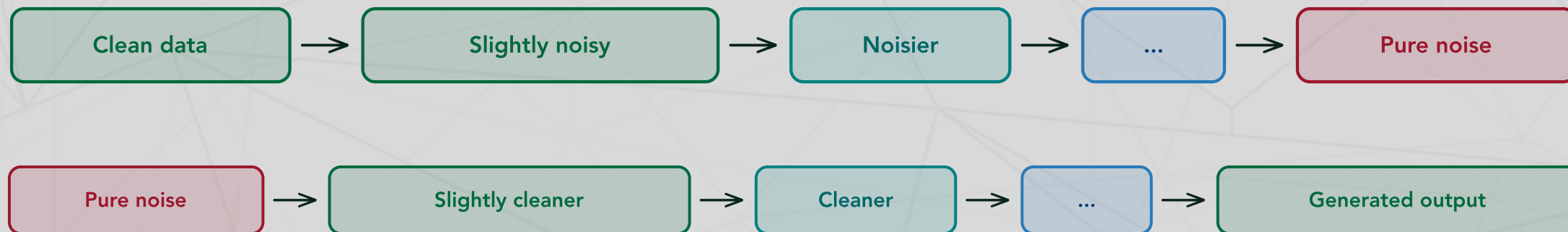
The diffusion framework: big picture

Two processes, one framework

Every diffusion model has two processes:

1. **Forward process** (corruption): Gradually destroy information in the data until nothing remains but noise
2. **Reverse process** (generation): Learn to undo the corruption, step by step, recovering the original data from noise

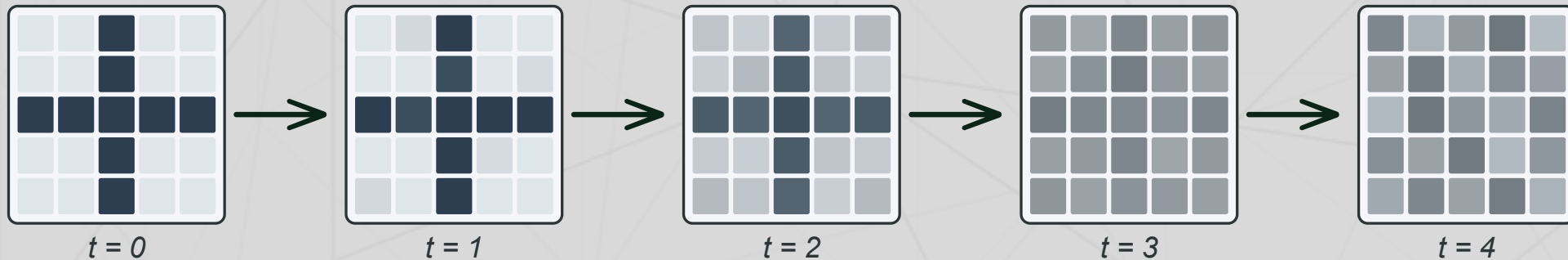
The key insight from Sohl-Dickstein et al. (2015): if each corruption step is **small enough**, then each reversal step can be learned by a neural network.



Diffusion was born in image space

The visual origin

Diffusion models were originally developed for **image generation** (Sohl-Dickstein et al., 2015; Ho et al., 2020). The idea is intuitive with images: start with a clean picture and gradually corrupt it by adding random noise until the image is unrecognizable. Then train a neural network to reverse the process.



From images to text

This visual intuition made diffusion wildly successful for image generation (DALL-E, Stable Diffusion, Midjourney — covered in Lectures 22–23). But how do we extend this to **text**, where "adding a little noise to a word" is meaningless? We'll answer this after establishing the mathematical framework.

Continuous diffusion: the foundation

Adding Gaussian noise (Ho et al., 2020)

For continuous data (images, audio, embeddings), the forward process adds Gaussian noise at each timestep $t = 1, \dots, T$:

$$\mathbf{x}_t \sim \mathcal{N}\left(\sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I}\right)$$

where β_t is a small noise amount at step t . After $T \approx 1000$ steps, the original signal is completely destroyed — $\mathbf{x}_T$ is indistinguishable from pure Gaussian noise.

Closed-form shortcut

We can jump directly to any timestep. Define $\bar{a}_t = \prod_{s=1}^t (1 - \beta_s)$:

$$\mathbf{x}_t = \sqrt{\bar{a}_t} \mathbf{x}_0 + \sqrt{1 - \bar{a}_t} \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

This is essential for efficient training — no need to iterate through all t steps.

Continuous diffusion: learning to reverse

The DDPM training objective

A neural network $\epsilon_\theta(\mathbf{x}_t, t)$ is trained to **predict the noise** that was added at each step. The loss is simply mean squared error (Ho et al., 2020):

$$\mathcal{L} = \mathbb{E}_{t, \mathbf{x}_0, \epsilon} \left[\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|^2 \right]$$

To generate, start from pure noise $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and iteratively subtract the predicted noise, step by step, until clean data $\mathbf{x}_0$ emerges.

Why we're covering this...

Lectures 22 and 23 build on this continuous framework (latent diffusion, classifier-free guidance, DiT, Stable Diffusion, Sora). Understanding Gaussian diffusion provides the conceptual foundation — but **text is not continuous**. Tokens are discrete symbols. So how do we adapt diffusion for language?

The problem with text

Tokens aren't pixels

Adding Gaussian noise to a sentence doesn't work:

- Pixel values are continuous numbers (0–255) — small perturbations are meaningful
- Token IDs are discrete symbols — "adding 0.1 to the token 'cat'" is meaningless
- There's no natural metric space where "cat" is close to "car" but far from "democracy"

Two solutions emerged

1. **Continuous embedding approach** (Diffusion-LM; Li et al., 2022): Map tokens to continuous embeddings, run Gaussian diffusion there, then round back to tokens
2. **Discrete diffusion** (D3PM; Austin et al., 2021): Replace Gaussian noise with **discrete corruption** — randomly changing tokens to other tokens or to **[MASK]**

The discrete approach turns out to be simpler, more effective, and connects beautifully to something you already know.

Discrete diffusion: replacing noise with masking

The absorbing state forward process

Instead of adding Gaussian noise, the forward process **randomly masks tokens**:

- At each step, each unmasked token has a small probability of being replaced with `[MASK]`
- As t increases, more tokens become masked
- At $t = T$: **every token is masked** — the sequence is pure noise (all `[MASK]`)
- At $t = 0$: the original clean text



Why 'absorbing state'?

Once a token becomes `[MASK]`, it stays masked. The `[MASK]` state **absorbs** tokens — they can't escape. This is the simplest discrete corruption: there's only one type of noise, and...you've seen it before!

Does this remind you of anything?

BERT is (almost) a diffusion model!

Recall from Lecture 18: BERT trains by masking 15% of tokens and predicting the originals. Discrete diffusion does the **same thing** — but with a key generalization:

	BERT (Lecture 18)	Discrete diffusion
Mask rate	Select 15%, then 80/10/10	Varies from 0% to 100% over a schedule
Prediction	One-shot: predict all masks at once	Iterative: unmask a few tokens at a time
Training	Single forward pass per example	Sample random mask rate t , predict masked tokens
Generation	Not designed for generation	Built for generation: start at 100% masked, iteratively unmask

The insight

Discrete diffusion **generalizes** masked language modeling into a generation framework. BERT is a special case — a single-step diffusion model at a fixed noise level.

MDLM: masked diffusion language models

Sahoo et al. (2024, NeurIPS)

MDLM formalizes the connection between masked language modeling and diffusion. The key ingredients:

1. **Forward process:** A continuous-time masking schedule $\gamma(t)$ that specifies the probability a token is masked at time $t \in [0, 1]$. At $t = 0$, nothing is masked; at $t = 1$, everything is.
2. **Reverse process:** A transformer predicts the identity of each masked token, conditioned on all visible tokens and the current noise level t .
3. **Training:** Sample a random time t , mask tokens according to $\gamma(t)$, and train the model to predict the masked tokens — exactly like BERT, but at every mask rate.

Rao-Blackwellization

Rao-Blackwellization is a statistical technique: when estimating something, if you can **compute part of the answer exactly** instead of approximating it, your overall estimate becomes more precise. In diffusion training, this means replacing some approximation steps with exact calculations — leading to less noisy gradients and faster learning.

Why MDLM works so well

MDLM uses a **Rao-Blackwellized** training objective that reduces variance compared to naïve discrete diffusion. In practice, this means the model trains efficiently with the same architecture as BERT — no special modifications needed. On text benchmarks, MDLM matches autoregressive models of the same size.

How text diffusion generates

The reverse process: iterative unmasking

To generate text with MDLM, we reverse the masking:

1. Start with a sequence of all `[MASK]` tokens (length N)
2. At each step, the model predicts a probability distribution over the vocabulary for each masked position
3. **Unmask** a subset of positions by sampling from these distributions
4. Repeat until all positions are unmasked



Bidirectional context at every step

Unlike GPT, which can only see tokens to the **left**, the diffusion model sees **all unmasked tokens** when predicting each mask — regardless of position. This means the model can use "mat" to help predict "cat" and vice versa. The generation order is not fixed — the model decides what to fill in first.

What does the model learn to do first?

Emergence of generation order

When generating text via iterative unmasking, models don't unmask uniformly. They develop preferences:

- **High-confidence tokens first:** Function words ("the", "is", "of") and predictable tokens tend to be unmasked early
- **Content words later:** Nouns, verbs, and less predictable tokens come after the "scaffolding" is in place
- **Context-dependent tokens last:** Words that depend heavily on surrounding context wait until that context is available

A linguistic insight

This mirrors how humans plan sentences — we often have the **gist** (content words) and **structure** (function words) before the exact phrasing. Diffusion models discover a similar strategy: build the scaffolding first, then fill in the details. This is fundamentally different from GPT's strict left-to-right constraint.

D3PM: the general framework

Austin et al. (2021, NeurIPS)

D3PM (Discrete Denoising Diffusion Probabilistic Models) provides the theoretical foundation for all discrete diffusion. Instead of Gaussian noise, D3PM uses **transition matrices** $\mathbf{Q}_t$ that define how tokens corrupt at each step:

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \text{Cat}(\mathbf{x}_t; \mathbf{x}_{t-1} \mathbf{Q}_t)$$

where Cat is the categorical distribution and $\mathbf{Q}_t$ specifies the probability of each token transitioning to any other token.

Three types of discrete noise

Noise type	How it works	Intuition
Uniform	Any token $\rightarrow$ any random token	Like replacing letters with random ones
Absorbing	Any token $\rightarrow$ [MASK] only	Like erasing letters one by one
Token similarity	Token $\rightarrow$ similar token	Like introducing typos

MDLM uses **absorbing** noise because it's the simplest and most effective for text.

Scaling up: LLaDA

Nie et al. (2025): Large Language Diffusion with Assistant

LLaDA asks: can discrete diffusion compete with autoregressive models **at scale**? They trained an 8 billion parameter masked diffusion model on 2.3 trillion tokens and found:

- On MMLU, LLaDA-8B (66.6) matches LLaMA 3-8B (66.2)
- On GSM8K (math reasoning), LLaDA-8B (74.0) substantially outperforms LLaMA 3-8B (52.8)
- LLaDA handles instruction following, reasoning, and long-form generation
- It naturally solves the **reversal curse** — since it doesn't generate left-to-right, it can answer "Who is A's mother?" even when trained on "B is the mother of A"

Why this matters

For years, the assumption was that autoregressive generation was the only way to build competitive language models. LLaDA disproves this. Diffusion-based LLMs are a viable alternative paradigm — not just a research curiosity, but a practical one at the 8B parameter scale.

The Diffusion-LM approach: embeddings as a bridge

Li et al. (2022, NeurIPS)

Before discrete diffusion matured, Diffusion-LM tried a different approach: run Gaussian diffusion in **continuous embedding space**:

1. Map each token to its embedding vector (e.g., 768-dim)
2. Run standard Gaussian diffusion on these embeddings
3. At generation time, round each denoised embedding back to the **nearest vocabulary token**

Tradeoffs

	Diffusion-LM (continuous)	MDLM / D3PM (discrete)
Noise type	Gaussian in embedding space	Masking or token swaps
Rounding step	Required (introduces errors)	Not needed
Controllability	Excellent — gradient-based guidance	Good — conditional masking
Scalability	Limited by rounding artifacts	Scales to 8B+ parameters

Discrete methods have won out for pure text generation, but continuous approaches remain valuable for **controllable generation** — e.g., guiding text toward specific sentiment, topics, or structural constraints.

Intuition: why does iterative refinement work?

Three intuitions for text diffusion

- 1. The editing analogy:** A first draft with blanks is easier to improve than starting from scratch. Each unmasking step gives the model more context, making the remaining predictions easier.
- 2. The jigsaw puzzle analogy:** In a jigsaw puzzle, each piece you place constrains where other pieces go. Similarly, each unmasked token constrains the remaining tokens — the problem gets easier as you go.
- 3. The ensemble effect:** At each step, the model makes predictions using **bidirectional context** from all visible tokens. Early steps leverage global structure; later steps leverage local details. This multi-scale reasoning is hard for autoregressive models.

The deep insight

Diffusion works because it **factorizes a hard problem** (generating a full sequence from nothing) into a sequence of **easier problems** (predicting a few tokens given most of the context). Each step is essentially a fill-in-the-blank task — something neural networks are very good at.

Practical advantages of text diffusion

What diffusion can do that autoregressive models can't (easily)

Capability	Autoregressive	Text diffusion
Infilling: Fill in a gap mid-sentence	Requires special fine-tuning	Native — just mask the gap
Iterative editing: Refine parts of generated text	Must regenerate from the edit point	Re-mask and re-denoise locally
Length control: Generate exactly N tokens	Hard — models tend to over/under-generate	Natural — initialize N masks
Parallel decoding: Generate multiple tokens simultaneously	Sequential by definition	Unmask multiple positions per step
Bidirectional coherence: Ensure beginning matches end	Can't look ahead	Sees all unmasked positions

The tradeoff

Text diffusion typically requires 10–100 refinement steps, each involving a full forward pass. Autoregressive generation requires N steps (one per token) but each step is faster. For short sequences, autoregressive wins on speed. For tasks requiring bidirectional coherence or editing, diffusion wins on quality.

Current limitations

Where text diffusion still falls short

1. **Long-range coherence:** Autoregressive models maintain a running context that naturally ensures consistency. Diffusion models must learn long-range dependencies through the iterative process, which can fail for very long documents.
2. **Sampling speed:** Even with optimized schedules, generating text with diffusion is slower than autoregressive generation with KV-caching for most practical sequence lengths.
3. **Ecosystem maturity:** Autoregressive models have years of tooling — RLHF, DPO, KV-caching, speculative decoding. Diffusion-based LLMs are catching up but lack this infrastructure.
4. **Evaluation:** Perplexity (the standard LM metric) doesn't directly apply to diffusion models, making fair comparison difficult.

Definitions

- **RLHF** (Lecture 16): Reinforcement Learning from Human Feedback — humans rank model outputs, and the model is trained to prefer higher-ranked responses
- **DPO** (Direct Preference Optimization): A simpler alternative to RLHF that skips the reward model and directly optimizes the language model on human preference pairs
- **KV-caching:** During autoregressive generation, previously computed key/value vectors are stored and reused so each new token only requires one forward pass through the new position — not the entire sequence

Take-home messages

Think about it...

- BERT's masked prediction training (Lecture 18) — which seemed like just a pretraining trick — turns out to be the **foundation of an entire generation paradigm**.
- The idea of **iterative refinement** from noise is a powerful principle that extends beyond text to images, audio, and video.
- Because the same diffusion framework applies to both **continuous** data (images, audio, video) and **discrete** data (text), it provides a natural foundation for **multimodal models** that generate across modalities — as we'll see in Lectures 22–23.

Further reading

Further reading

Sohl-Dickstein et al. (2015, *ICML*) "Deep Unsupervised Learning using Nonequilibrium Thermodynamics" — The original diffusion model paper, grounded in statistical physics.

Ho, Jain & Abbeel (2020, *NeurIPS*) "Denoising Diffusion Probabilistic Models" — Made diffusion practical with the simplified noise-prediction objective.

Austin et al. (2021, *NeurIPS*) "Structured Denoising Diffusion Models in Discrete State-Spaces" — D3PM: the theoretical foundation for all discrete diffusion.

Li et al. (2022, *NeurIPS*) "Diffusion-LM Improves Controllable Text Generation" — Continuous embedding approach to text diffusion with gradient-based control.

Lou et al. (2024, *ICML*) "Discrete Diffusion Modeling by Estimating the Ratios of the Data Distribution" — SEDD: score matching for discrete spaces (Best Paper).

Sahoo et al. (2024, *NeurIPS*) "Simple and Effective Masked Diffusion Language Models" — MDLM: connecting BERT-style masking to diffusion.

Nie et al. (2025, *arXiv*) "Large Language Diffusion Models" — LLaDA: 8B-parameter diffusion LLM competitive with LLaMA 3.

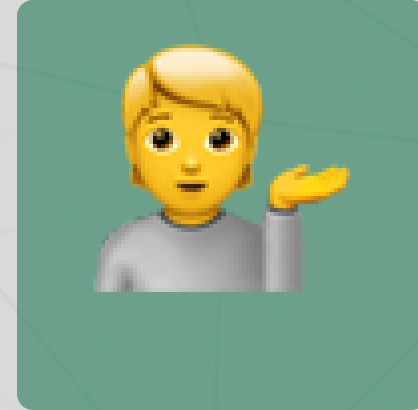
Questions?



Email



Discord



Office hours

Up next...

Diffusion model extensions: latent diffusion, classifier-free guidance, and the Diffusion Transformer